

Agent-Based Infrastructures for Diachronic Digital Humanities

Cohort

Agents for retrieving, comparing and documenting textual evidence

Shu-Kai Hsieh¹² · Da-Chen Lian¹² · Chunki Lim¹ · Lang-Ching Yeh¹

¹ Graduate Institute of Linguistics, National Taiwan University, Taipei, Taiwan

² Humanistic-AI lab, College of Liberal Arts, National Taiwan University, Taipei, Taiwan

Acknowledgement

Thank you, Michael Radich

Texts

A modified, selected
CBETA-derived corpus

Catalogue

Translator labels and
mixed historical groups

Vocabulary

Curated strings for
Dharmarakṣa research

Cohort builds the profiles and calculates the rankings.

Why we built Cohort

Let researchers direct agents and inspect the evidence behind their proposals.

Today's presentation

Use Buddhist texts to show the functions, where results can mislead, and what the researcher decides.

We are demonstrating research tools, not reporting a new translator attribution.

Agent support for diachronic research

Study texts and their relationships across time.

Corpus retrieval

Find relevant texts and passages.

Semantic analysis

Find passages with related content.

Alignment

Locate matching wording and positions.

Interpretation

Propose explanations and alternatives.

Provenance

Keep sources and checks attached to proposals.

Researcher decisions

Inspect the evidence; accept or reject proposals.

Functions

Corpus

Find passages and read them in context.

Vocabulary comparison

Compare a text's wording with texts labelled by translator.

Inquiry

Give agents a research question and inspect their actions.

Findings

Read the agents' proposals, citations and checks.

Graph

Follow links between sources, proposals and decisions.

Example research questions

Two Lotus translations

Dharmarakṣa 竺法護 · T0263
Kumārajīva 鳩摩羅什 · T0262

Can we find related passages
when the wording differs?

Text and later commentary

Yin chi ru jing 陰持入經 · T0603
Body, mind and senses

Could shared wording mislead us
about who translated this text?

Known relationships let us inspect what the tools get right and wrong.

Find passages and read them in context.

The screenshot displays the COHORT Corpus interface. At the top, there's a navigation bar with tabs: Graph, Findings, Corpus (selected), Vocabulary comparison, and Inquiry. Below this, the 'Corpus' section is active. It features a search bar with 'Exact phrase' and 'Related passages' tabs. A message states: 'Find passages with similar content in other works. Start from an indexed text, not a typed research question.' Below this, it shows the 'Embedding model (from filename): mitra-qwen35-embedder' and statistics: '2,160 indexed texts/chapters • 54,616 passages • 400-character windows'. A section titled 'Model and search method' is expanded. The 'Indexed text' dropdown is set to 'Lotus Sūtra 正法華經 · Dharmarakṣa · catalogue remainder · T0263' and the 'Source position' is '4000'. A 'Find related passages' button is present. Below the search bar, there are tags for 'Lotus Sūtra · Dharmarakṣa', 'Body, mind and senses · T0603', and 'Maitreya · T0453'. The 'SELECTED PASSAGE' section shows the source text: 'LOTUS SŪTRA 正法華經 · DHARMARAKṢA · CATALOGUE REMAINDER · T0263-REST' and the source characters '4000–4400 (zero-based, end excluded)'. It also includes buttons for 'Previous passage' and 'Next passage'. The 'RELATED PASSAGES' section shows a list of candidate passages ranked by cosine similarity. The first result is '1. Lotus Sūtra 妙法蓮華經 · Kumārajīva · without Devadatta chapter · T0262-exDevadatta' with a cosine similarity of 0.865. The second result is '2. T0278-ex-earlier-parallels' with a cosine similarity of 0.856. Both results show their source characters and a note that the related passage is hidden in the screenshot.

Source passages hidden; controls and recorded outputs retained.

Find

Search exact wording or select Related passages.

Read

Selected passage on the left; candidates on the right.

Compare

Open shared wording and inspect source positions.

How related passages are found

mitra-qwen35-embedder

Selected passage

Use its stored vector

Cosine similarity



Compare with other works

Related passages



Rank candidates to read

Corpus → Related passages; agents can call `semantic_neighbors`.

Model name comes from the index filename; its exact revision is unrecorded.

Example: related content, little shared wording

Earlier Lotus translation

Dharmarakṣa · T0263-rest
Source position 4000

cosine 0.865



Later Lotus translation

Kumārajīva · T0262-exDevadatta
Source position 2800

Longest exact shared sequence: 3 Chinese characters

Shared wording is checked separately, with positions in both sources.

What vocabulary comparison calculates

Vocabulary list

Choose strings to count



Target text

Count those strings



Group profiles

Texts pooled by
catalogue label



Compare usage → rank the catalogue groups

13 usable groups in this dataset. The target's whole work is withheld.

Compare a text's wording with texts labelled by translator.

COHORT

beta

GraphFindingsCorpusVocabulary comparisonInquiry

ⓘ⚙

Analyze this view

Compare a text's wording

Compare short-string counts across corpus groups.

Try a text: Lotus Sūtra • Dharmarakṣa Yin chi ru jing 陰持入經 • T0603 Maitreya's descent 彌勒下生經 • T0453

Examples start with the curated strings and no additional exclusions.

► Change the selected text

► Vocabulary and method

Yin chi ru jing 陰持入經 • T0603

Catalogue classification: An Shigao 安世高 (ASg)

9,254 Han characters (12,012 code points) • 45 matched string occurrences • 35 different strings • vocabulary: Radich's curated strings (7,270 strings) • 1 unit withheld from the profiles

Highest-ranked comparison group	Next comparison group	Score difference
Other material before Dharmarakṣa 竺法護 (pre-Dhr-other)	An Shigao 安世高 (ASg)	+1.696
#11 of 21 in Radich's order	#1 of 21 in Radich's order	log-odds per distinct string; not comparable across texts

Rankings

Which group's string frequencies resemble this text?

Colours

Teal adds weight to A; rust adds weight to B.

Exclusions

Remove comparison material and calculate again.

T0603: remove the later commentary

With T1694

First: Other material before
Dharmarakṣa 竺法護

Margin over next group: 1.696

Without T1694

First: An Shigao 安世高

Margin over next group: 0.096



T1694 is the commentary on T0603. Its wording affects the ranking.

This agrees with the accepted attribution after exclusion; it does not establish one.

Give agents a research question and inspect their actions.

The screenshot shows the 'COHORT beta' application with a top navigation bar containing 'Graph', 'Findings', 'Corpus', 'Vocabulary comparison', and 'Inquiry' (which is highlighted). The 'Inquiry' modal is open, displaying the following steps and controls:

- Inquiry**
Choose an example or write a question to begin.
- 1. Choose or record a research question** (with a 'Cancel' button)
Choose an example, then edit and record it.
Three buttons are available: 'Find related passages: T0603', 'T0453: find a parallel', and 'Test recurring formulae'.
- Research question**
A text box contains: 'Which passages are related to the Yin chi ru jing (T0603), and what might explain the relationship?'.
- Research instructions**
A text box contains: 'Find related passages, compare shared wording, and consider quotation, commentary and recurring formulas. Report source positions and limitations.'.
- Record question**
A large purple button to save the question.
- 2. Start a new agent run**
Below this, it says 'Specify the approach, evidence to return and limits. These instructions are saved and sent to the agents.' and shows 'Saved questions (4)' with a 'Show hidden questions (3)' button.

Source passages hidden; controls and recorded outputs retained.

Question

What do you want to find out?

Instructions

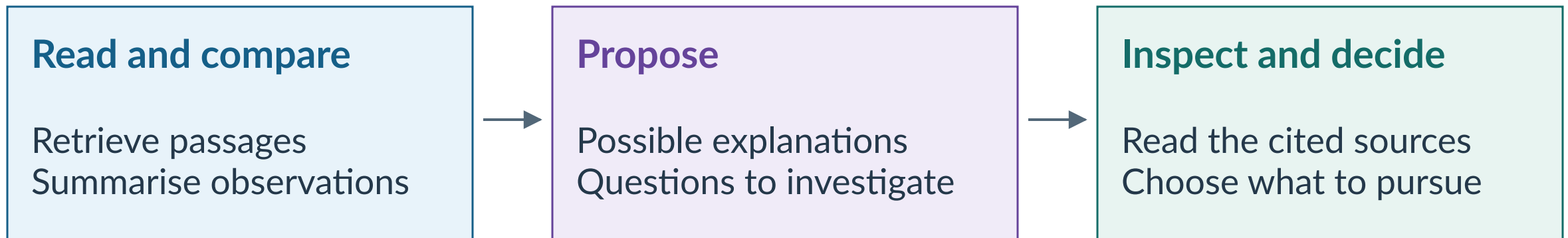
How should the worker investigate it?

Run results

Reopen the actions, inputs and outputs.

From passages to research proposals

Agents can compare material, suggest explanations and propose follow-up searches.



Graph

Keeps proposals linked to passages, checks and researcher decisions.

A citation that resolves can still support a weak interpretation.

Read the agents' proposals, citations and checks.

COHORT beta

Graph Findings Corpus Vocabulary comparison Inquiry

Hypotheses **8**

Claims and conjectures, newest first. Not ranked by confidence.

CONJECTURE The close relationship between Yin chi ru jing (T0603) and T1694 reflects textual dependence or shared formulaic stock — quotation, commentary, or reuse of the same base exposition — rather than independent composition; whether it also reflects a shared hand cannot be settled from this corpus alone.

ATTESTED A2 EXACT SPAN MATCHED 2 ATTESTING • 2 WITNESSES • INDEPENDENT PROSPECTIVE QUERY NOT YET RUN DOSSIER

CLAIM Yin chi ru jing (T0603) and T1694 reproduce the same extended wording, including the eighteen-base exposition and long doctrinal runs up to 279 characters, with 64.45% of T0603's ten-character strings found in T1694.

ATTESTED A2 EXACT SPAN MATCHED 2 ATTESTING • 2 WITNESSES • INDEPENDENT

CONJECTURE The apparent similarity between T0603 and Other material before Dharmarakṣa 竺法護 (pre-Dhr-other) translator vocabulary is driven by repeated material shared with T1694: once T1694 is withheld from the comparison profiles, the leaning for T0603 collapses to no meaningful distinction between An Shigao 安世高 (ASg) and Other material before Dharmarakṣa 竺法護 (pre-Dhr-other).

ATTESTED A2 EXACT SPAN MATCHED 2 ATTESTING • 2 WITNESSES • INDEPENDENT PROSPECTIVE QUERY NOT YET RUN DOSSIER

CLAIM Yin chi ru jing (T0603) reproduces wording also found in T1694 at length, including a 128-character run enumerating the eighteen bases and a 279-character run, with 64.45% of T0603's ten-character strings found in T1694.

ATTESTED A2 EXACT SPAN MATCHED 2 ATTESTING • 2 WITNESSES • INDEPENDENT

CLAIM T0603 (陰持入經) and T1694 reproduce the same extended wording verbatim, including a 128-character run beginning 四耳五聲六識 and a 279-character run on non-extinction/purity; 64.5% of T0603's ten-character strings occur in T1694.

ATTESTED A2 EXACT SPAN MATCHED 2 ATTESTING • 2 WITNESSES • INDEPENDENT

CONJECTURE The Other material before Dharmarakṣa 竺法護 (pre-Dhr-other) leaning of T0603 (陰持入經) under Radich's vocabulary depends on T1694 sharing its wording: with T1694 withheld the leaning disappears (An Shigao 安世高 (ASg) ahead by margin 0.096, gap 3.4, i.e. no evidence either way), so the corpus as configured cannot settle whether An Shigao translated T0603.

ATTESTED A2 EXACT SPAN MATCHED 2 ATTESTING • 2 WITNESSES • INDEPENDENT PROSPECTIVE QUERY NOT YET RUN DOSSIER

Source passages hidden; controls and recorded outputs retained.

Proposal

A claim or a conjecture from the worker.

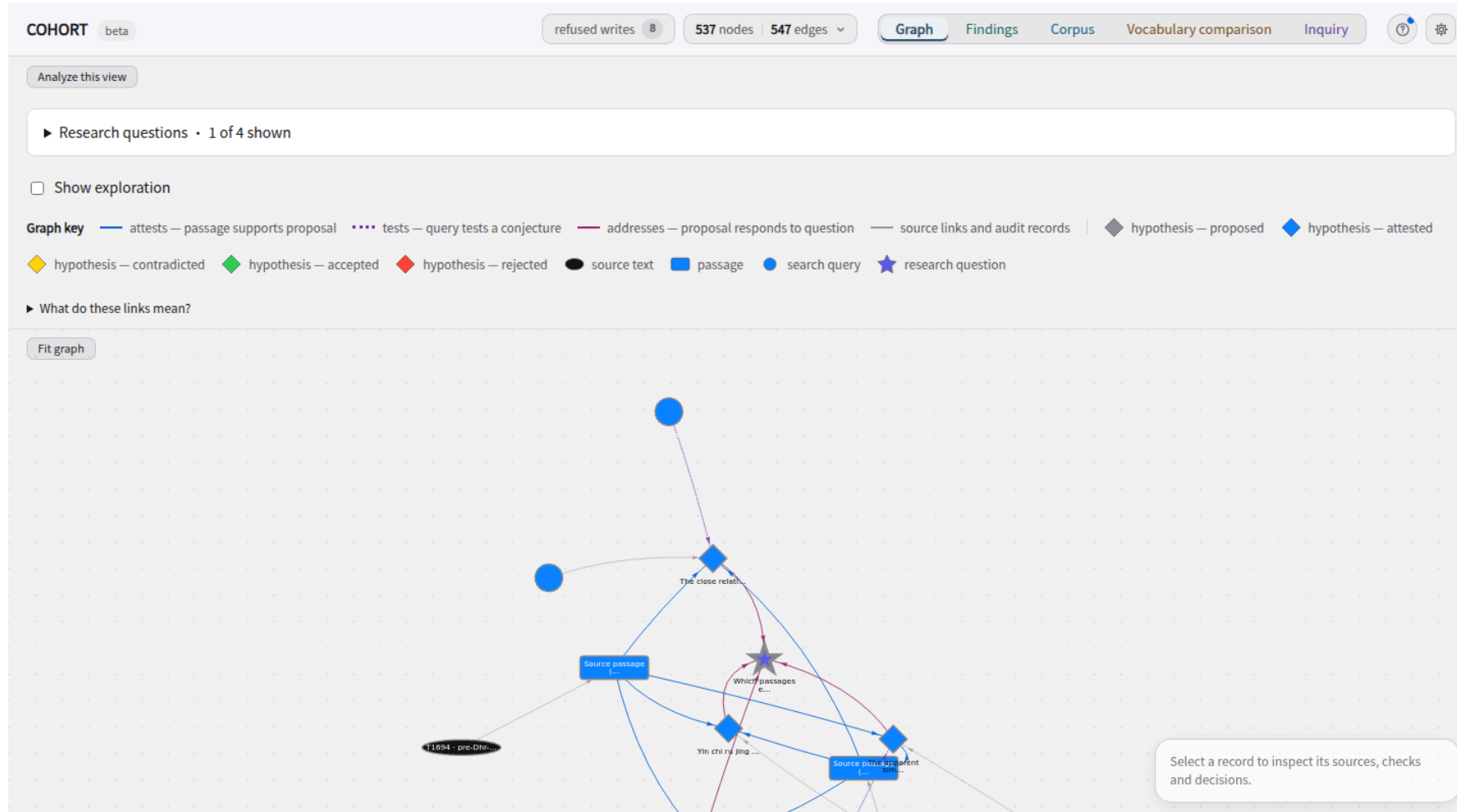
Sources

Read its citations and recorded checks.

Alternatives

Inspect what else could explain the observation.

Follow links between sources, proposals and decisions.



Source passages hidden; controls and recorded outputs retained.

Records

Sources, passages, proposals and questions.

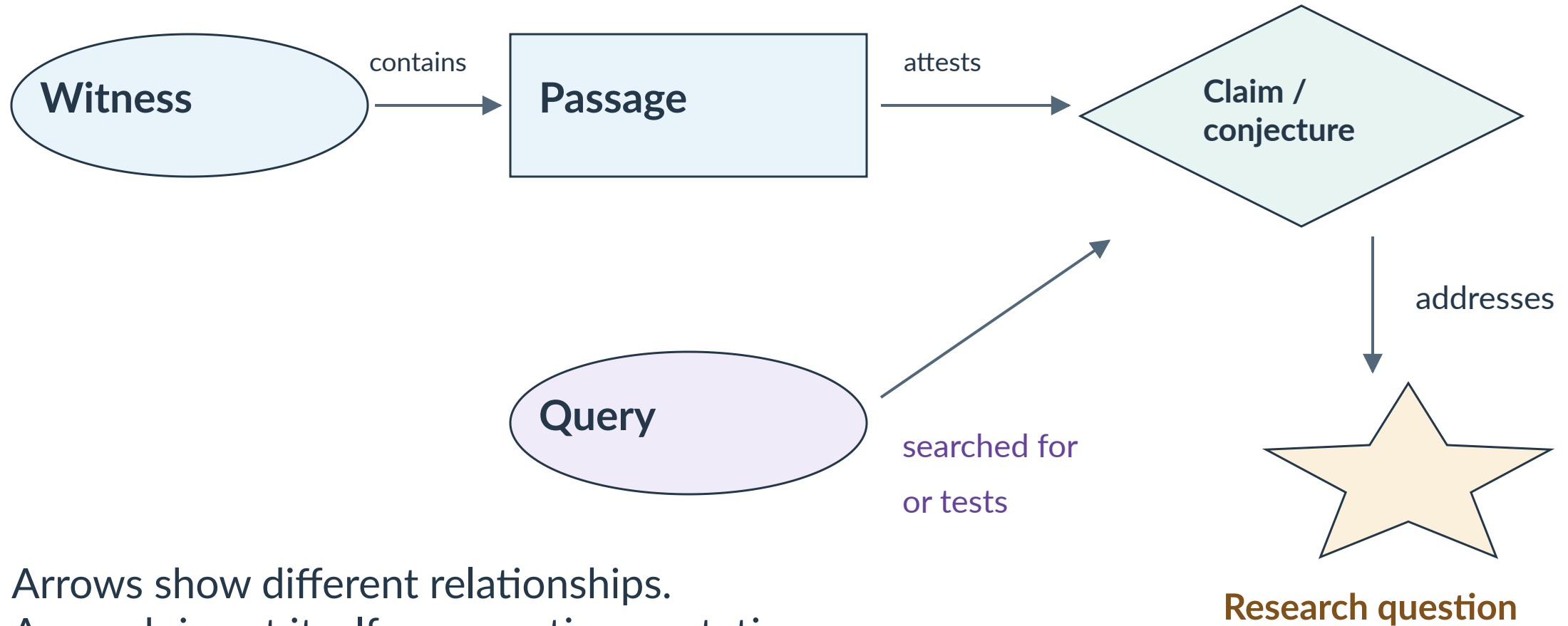
Links

Read the relationship label on each arrow.

Exploration

Show recorded tool activity beyond cited evidence.

Why a query points to a proposal



Checks and researcher decisions

Span verified

The quotation resolves at its source position.

Review recorded

Inspect the reviewer's verdict and evidence.

Researcher decides

Accept or reject an eligible record.

Accepted passage ≠ accepted interpretation

Decisions stay in the record. The source text is unchanged.

Cohort

- 1 Find** Lotus passages and shared wording
- 2 Compare** T0603 with and without its commentary
- 3 Inspect** Saved actions, proposal, citations and checks

Results and limits

Useful outputs

Lotus: a related passage with little exact shared wording.

T0603: exclusion exposes the commentary's effect on the ranking.

Proposals, sources and checks can be inspected together.

Observed limits

Kumārajīva's Lotus wrongly ranks Dharmakṣema first in vocabulary.

One known-answer example cannot validate attribution.

Weak tests and reviewer errors still require researcher scrutiny.